Does Mamba Earn Its Place?: script

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

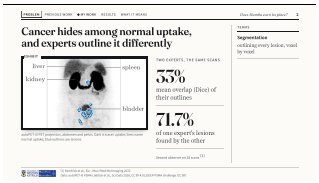
13 main slides · 1796 words · 13:37 spoken at 135 words a minute (including 19 s of clips without narration) · target 13:39



`Slide 1 · 20 s · ends at 0:20 · 43 words`

1. Title. Say hello and credit the supervisors.

Hi everyone, I'm James. My honours project is about Mamba, a newer kind of neural network layer. I asked whether it earns its place in a network that outlines prostate cancer in whole-body PET scans. I'm supervised by Jake Kendrick and Mubashar Hassan.

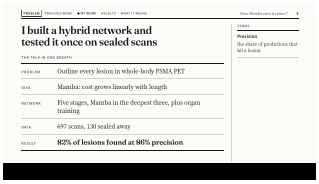


`Slide 2 · 62 s · ends at 1:22 · 140 words`

Cancer hides among normal uptake, and experts outline it differently

1. The exhibit, cropped to abdomen and pelvis: normal uptake labelled, lesions outlined in blue.
2. Click: the two observer figures appear.

Here's a PSMA PET scan of the abdomen and pelvis. PSMA is a protein on prostate cancer cells, and the scan's tracer sticks to it. PET/CT pairs a map of that tracer with a CT of the anatomy. This is the PET half, so darker means more tracer. `POINT AT THE BLUE OUTLINES` The outlined spots are lesions, separate deposits of cancer. But healthy tissue lights up too, like the kidneys and the bladder. Where the lesions are, and how much disease there is, both bear on treatment. So we want every lesion outlined, which is called segmentation. By hand it's slow, and experts don't agree. `CLICK` When a second expert outlined 28 of the same scans, the two sets of outlines overlapped with a mean Dice of only 33%. Dice measures how much two outlines share. The second found just 71.7% of the first's lesions. `(cut if behind)`

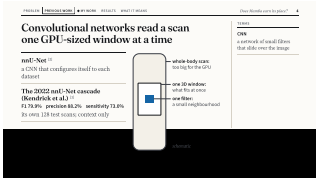


`Slide 3 · 30 s · ends at 1:52 · 68 words`

I built a hybrid network and tested it once on sealed scans

1. The abstract in one breath; no clicks.

I built a network that's convolutional in its shallow stages, with Mamba blocks in the deepest three. A side task teaches it organs during training. Of 697 public scans, I sealed 130 away for one final test. There it found 82% of lesions, which is its sensitivity. And 86% of its predictions hit a real lesion, which is its precision. Then come four predictions, each with a verdict. `(cut if behind)`

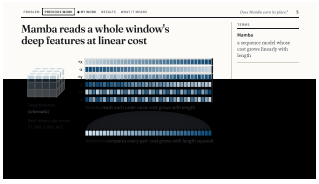


`Slide 4 · 65 s · ends at 2:57 · 151 words`

Convolutional networks read a scan one GPU-sized window at a time

1. Previous work: nnU-Net, the cascade figures as context, PSMASegmentator.
2. Click: the nested scan, window and filter schematic.

The usual starting point is nnU-Net, a convolutional neural network, or CNN, that configures itself to each dataset. A CNN is built from small filters that slide over the image. In 2022, Kendrick and colleagues built a two-stage nnU-Net cascade for whole-body PSMA PET. On their own 128 test scans it reached an F1 of 79.9%, a score that balances lesions found against false alarms. Its precision was high, but it found only 73.0% of lesions. Those figures are context only, because that test set isn't available to me. The model I compare against is PSMASegmentator, a later nnU-Net trained on 1,035 scans. `CLICK` `POINT AT THE NESTED BOXES` Here's the catch. A whole-body scan doesn't fit in GPU memory, the memory on the graphics card. So a CNN reads the scan one 3D window at a time, and a bigger window leaves room for fewer windows per training step. Each filter sees only a small neighbourhood. `(cut if behind)`

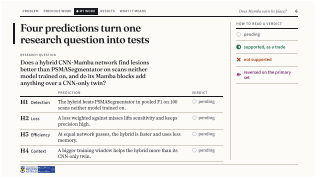


`Slide 5 · 75 s · ends at 4:12 · 170 words`

Mamba reads a whole window's deep features at linear cost

1. Mamba defined; the deep feature block.
2. Click: the block flattens into six token lanes.
3. Click: each lane is read once, left to right.
4. Click: attention web over the same tokens.
5. Click: the U-Mamba warning, so a twin.

My project bets on Mamba, a sequence model whose cost grows linearly with sequence length. `CLICK` To use it on a scan, the network takes a 3D block of features from deep inside itself. It flattens that block into a line of tokens, and a token is one cell of the block. A sequence only runs one way, so each Mamba block reads the same tokens in six orders, forwards and backwards along each axis. `CLICK` Each order is read once, start to finish, carrying a small memory of fixed size. That's why the cost grows linearly. `CLICK` Attention, the layer inside transformers, compares every token with every other token, so its cost grows with the square of the length. My deepest stages hold up to 27,648 tokens per order. `(cut if behind)` Mamba still only sees one window, though, never the whole body. `CLICK` When Isensee and colleagues retested U-Mamba against a copy with its Mamba layers switched off, those layers had no effect. So a fair claim for Mamba needs a twin network without them.

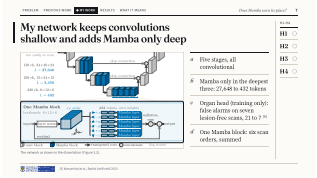


Slide 6 · 45 s · ends at 4:57 · 100 words

Four predictions turn one research question into tests

1. The research question, large; the table is hidden.
2. Click: the four predictions, all pending; the scoreboard is born.

My question has two halves. `PAUSE` First, does the hybrid find lesions better than PSMASegmentator on scans neither model has trained on? Second, if it does, is Mamba the reason? `CLICK` I turned that into four predictions. H1 is the head-to-head on unseen scans. H2 and H3 test two claims from my research proposal, about the loss function and about speed and memory. H4 is the heart of the bet. If Mamba really adds context, a bigger training window should help the hybrid more than its twin. For now all four are pending. `(cut if behind)` The margin column fills in as the evidence arrives.

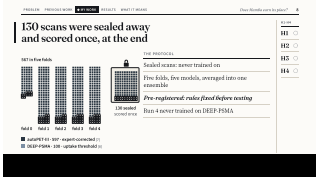


Slide 7 · 62 s · ends at 5:59 · 138 words

My network keeps convolutions shallow and adds Mamba only deep

1. Option F, whole figure.
2. Click: zoom onto the three deep stages.
3. Click: zoom onto the training-only organ head; say the organ result.
4. Click: zoom onto one Mamba block.

Here's what I built. `POINT AT THE FIGURE` It's a five-stage encoder-decoder. The encoder shrinks the scan while it learns features, and the decoder builds the outline back up at full resolution. `CLICK` Every stage has convolutions, and the deepest three also get Mamba blocks, because that's where the flattened sequence is short. At the shallow stages it would be at least 221,184 tokens per direction. Convolutions handle that fine texture more cheaply anyway. `(cut if behind)` `CLICK` The dashed box is an organ head. During training it learns 24 anatomical structures from automatic labels, so the network learns where normal uptake lives. It's never used when the model predicts. My third development run raised false alarms on all seven lesion-free validation scans, and adding this head cut them from 21 to 7. `CLICK` Zoom in, and one Mamba block is the six-order scan from before, outputs summed.

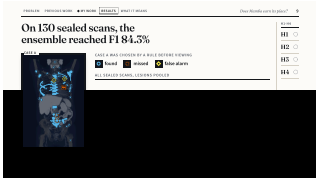


Slide 8 · 68 s · ends at 7:07 · 154 words

130 scans were sealed away and scored once, at the end

1. Two cohorts as tiles.
2. Click: the sealed test set moves into the vault.
3. Click: the pool splits into five folds.
4. Click: pre-registered.
5. Click: DEEP-PSMA tiles outlined; name run four and run six.

Now, how I tested it. I used two public cohorts, 697 scans in all. `POINT AT THE TILES` They label lesions differently. autoPET-III outlines were corrected slice by slice by an expert. DEEP-PSMA marks every voxel, or 3D pixel, at or above a fixed uptake threshold, with normal uptake painted out by a physician. `CLICK` First I sealed away 130 scans. Nothing trained on them, and I scored them once. `CLICK` The other 567 went into five-fold cross-validation. That means training five models, each with a different fifth held out for checking, then averaging all five into one ensemble. `CLICK` Before scoring the test set, I fixed the lesion cut-off, the smallest blob I'd keep and the results I'd report. That's what pre-registered means here. `CLICK` Along the way I trained numbered development runs. Run 4 is an early single hybrid that trained on autoPET only, so all 100 DEEP-PSMA scans are unseen by it. PSMASegmentator never trained on those scans either. `(cut if behind)`

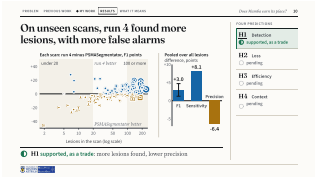


Slide 9 · 60 s · ends at 8:07 · 124 words

On 130 sealed scans, the ensemble reached F1 84.3%

1. Case A clip, cropped to the scan; pause while the rings trace on.
2. Click: the pooled sealed-test numbers.

So, does it work? This is one sealed scan, case A. I picked it by a rule, as the strongest of the larger autoPET scans, before looking at any images. `CLIP PLAYS, 8.8 S, PAUSE WHILE THE RINGS TRACE ON` Blue lesions were found, orange ones were missed, and yellow marks are false alarms. A lesion counts as found if any predicted voxel touches it. `CLICK` Across all 130 sealed scans, with lesions pooled, the ensemble's F1 was 84.3%. Sensitivity is the share of lesions found, and precision is the share of predictions that hit a lesion. Dice is lower, at 59.6%, because it needs the outlines to match voxel by voxel. These scans come from the cohorts it trained on, so this result is in-domain. Pooled means scans with lots of disease count most. `(cut if behind)`

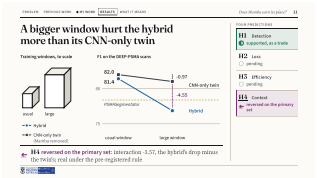


`Slide 10 · 85 s · ends at 9:32 · 187 words`

On unseen scans, run 4 found more lesions, with more false alarms

1. The underdog table.
2. Click: one hundred scans as dots, scattered by lesion count.
3. Click: dots grow by lesion count; the pooled differences rise in their own panel. Do not call the bars a weighted mean.
4. Click: H1 turns to supported, as a trade.

H1 is the fair test. PSMASegmentator trained on every autoPET scan in my test set, so I can't score it there. Instead I used the 100 DEEP-PSMA scans that neither model trained on, and run 4 rather than my final model. `POINT AT THE TABLE` On paper run 4 is the underdog. It had fewer training scans, one model instead of five, and fewer parameters. It also skipped test-time augmentation, where each scan is run again mirrored. `CLICK` Each dot is one scan: run 4's F1 minus PSMASegmentator's, against how many lesions the scan has. Scan by scan, there's no clear difference. Run 4 was better on 48 and worse on 50. It usually lost on scans with few lesions and usually won on scans with many. `CLICK` Pooled F1 counts lesions, so the big scans weigh the most. Pooled, run 4's F1 was 3.0 points higher, and its 95% interval, 1.2 to 4.7, stays above zero. It found more lesions, with sensitivity up 8.1 points, but its precision fell 6.4 points. `CLICK` So H1 is supported, but only as a trade: more lesions found for more false alarms. Mamba gets no credit yet. `(cut if behind)`

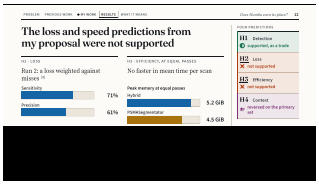


`Slide 11 · 92 s · ends at 11:04 · 207 words`

A bigger window hurt the hybrid more than its CNN-only twin

1. The design: two windows to scale, hybrid and twin.
2. Click: usual-window points, both level.
3. Click: the large window; both drop, the hybrid far more.
4. Click: the interaction; H4 turns to reversed on the primary set.

So is Mamba the reason? I trained a twin, the same network with every Mamba block removed. This is an ablation: take one part away and see what changes. `POINT AT THE WINDOWS` Each network was trained at the usual window and at a large one, with 4.7 times the voxels. H4 has its own pre-registered test: I fixed the comparisons and the decision rule before scoring any of these new models. `CLICK` At the usual window, on the 100 DEEP-PSMA scans, the two are level. On the autoPET test scans the hybrid was slightly ahead, so the Mamba blocks add little at this window. Both sit above PSMASegmentator, though only run 4's gap was tested with an interval. `CLICK` `PAUSE` The twin barely moves, losing 0.97 of a point. The hybrid drops 4.55 points. `CLICK` The gap between those drops, 3.57 points, is the interaction, and under my rule it's real. Its interval excludes zero and it's bigger than 1.0 point. `(cut if behind)` So H4 is reversed on the primary set, DEEP-PSMA: the bigger window hurt the hybrid more than its twin. On autoPET the difference was not clear. One possible reason is sequence length: at the large window the third stage reads 129,024 tokens, against 27,648, through the same small memory. I didn't test why.



`Slide 12 · 63 s · ends at 12:07 · 143 words`

The loss and speed predictions from my proposal were not supported

1. H2 panel: run two sensitivity and precision.
2. Click: H2 turns to not supported.
3. Click: H3 panel: no faster in mean time; peak memory bars.
4. Click: H3 turns to not supported.

H2 and H3 came from my proposal. H2 was about the loss. Run 2 used a loss that punishes missed lesions more than false alarms. `POINT AT THE LEFT PANEL` It found 71% of lesions, but its precision fell to 61%, and no threshold brought it back. `CLICK` So H2 is not supported, though it was a partial test: one run, on 57 validation scans, under an older scoring rule. `CLICK` H3 was about speed and memory. In their default setups, my ensemble took 18.1 s per scan against 43.8 s for PSMASegmentator. But that gap comes from PSMASegmentator's eight-pass test-time augmentation. The abstract's 32 against 63 seconds, from my home GPU, is the same effect. At equal passes the hybrid was no faster in mean time, and its peak memory was 5.2 GiB against 4.5 GiB. `CLICK` So H3 is not supported either. The hybrid is the smaller network, though. `(cut if behind)`

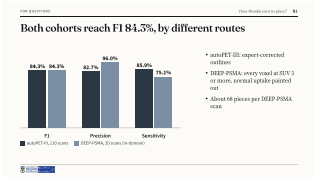


`Slide 13 · 92 s · ends at 13:39 · 171 words`

Run 4 beat PSMASegmentator in pooled F1, but Mamba takes no credit

1. The answer, with the full scoreboard.
2. Click: limitations.
3. Click: next step and the recorded demo; let it play.
4. Click: thank you and questions.

So, does Mamba earn its place? `POINT AT THE SCOREBOARD` On the 100 DEEP-PSMA scans neither model had seen, run 4, my single hybrid, found more lesions than PSMASegmentator, at lower precision. But on those scans the CNN-only twin did as well. The Mamba blocks added little, so I don't credit the gain to them, and a bigger window hurt the hybrid more. The part I'm most confident in is the test design: unseen scans, and a pre-registered twin comparison. `CLICK` Each cell of the twin experiment was a single training run, and the large window trained at batch size 1. The two cohorts label lesions differently, and only 8 sealed scans were lesion-free. The timing also covers just 20 scans. `(cut if behind)` `CLICK` The next step is to retrain and score the model on the Sir Charles Gairdner Hospital cohort my proposal named, where it's untested. The code will be released, so the UWA Medical Physics Research Group can do that. Here's a real run in my app, sped up 6x. `CLIP PLAYS, 18 S, LET IT RUN` `CLICK` Thank you. I'm happy to take questions.



Backup B1 · 76 words

Both cohorts reach F1 84.3%, by different routes

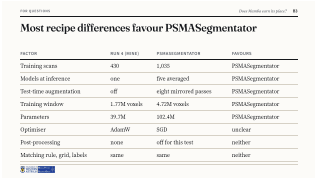
The two cohorts label lesions differently, so I report them separately. autoPET-III outlines were corrected slice by slice by an expert. DEEP-PSMA marks every voxel at SUV 3 or more, with normal uptake painted out, so a scan splits into about 68 small pieces. The likely reason for the gap: on DEEP-PSMA nearly everything the model predicts lands on disease, so precision is 96.0%, but it misses many small pieces. Both land on the same F1.



Backup B2 · 63 words

Cross-validation agreed with the sealed test

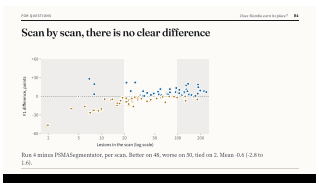
Before the test, each of the 567 pool scans was predicted by the fold model that never trained on it. That pooled F1 was 81.8%, with a fold-to-fold spread of 1.5 points, and it sits inside the test interval of 81.4 to 86.5. A single fold model scored 83.1 on average on the test set, so averaging the five added about 1.2 points.



Backup B3 · 70 words

Most recipe differences favour PSMASegmentator

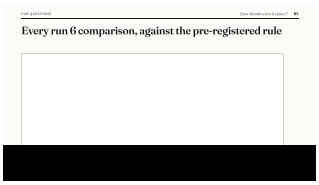
This is the honest list of what differs. PSMASegmentator trained on 1,035 scans against my 430, averages five models with eight mirrored passes, uses a bigger window, and has 102.4M parameters against 39.7M. It also has deep supervision and a misalignment augmentation that I did not port. Most factors favour it, so my gain is more likely to understate than overstate the recipe, but none was tested on run 4.



Backup B4 · 65 words

Scan by scan, there is no clear difference

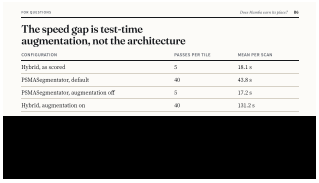
Each dot is one of the 100 DEEP-PSMA scans. Pooled F1 weighs lesion-heavy scans most, so the pooled gain comes from the right of this plot. Run 4 was below PSMASegmentator on 18 of the 22 scans with under 20 lesions and above it on 23 of the 28 with 100 or more. The mean per-scan difference was -0.6 points, with an interval spanning zero.



Backup B5 · 62 words

Every run 6 comparison, against the pre-registered rule

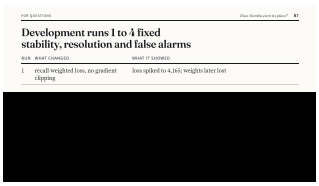
These are all five comparisons I committed to before scoring, from the dissertation. On DEEP-PSMA the interaction was -3.57 points, from -4.72 to -2.48, a real effect under the rule. On autoPET it was -1.62, no clear effect. The caveats: one training run per cell, the large window trained at batch size 1, and the large hybrid was selected at epoch 275.



Backup B6 · 73 words

The speed gap is test-time augmentation, not the architecture

Timed on one RTX 4090 over 20 test scans, each model already loaded. As scored, the hybrid took 18.1 s against 43.8 s for PSMASegmentator, but that gap is its eight-pass augmentation. At equal passes the hybrid was no faster in mean time and used 5.2 GiB of memory against 4.5 GiB. The abstract quoted 32 against 63 seconds from an earlier run on my home RTX 5070 Ti; the same caveat applies.



Backup B7 · 76 words

Development runs 1 to 4 fixed stability, resolution and false alarms

Run 1 had no gradient clipping and lost about three of its twelve hours to loss spikes. Run 2 added clipping, but the recall-weighted loss left precision at about 61%. Run 3 moved to the finer grid and Dice, then flagged every lesion-free validation scan. Run 4 added the organ head. Runs 2 to 4 were scored under an older one-to-one matcher; on identical run 4 predictions it gave 73.7 where the adopted rule gives 85.5.



Backup B8 · 64 words

Mamba sits only where the flattened sequence is short

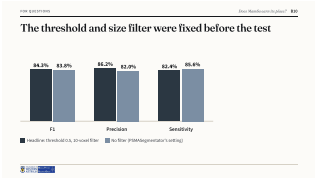
Each deeper stage halves the grid, so the sequence a Mamba layer would scan shrinks fast. The shallow stages would be at least 221,184 tokens per direction, and their features are fine texture a convolution handles cheaply. In the deep three, each block scans six directions and sums them. I never tested the alternatives, such as alternating directions across layers or a larger state.



Backup B9 · 68 words

Case B: both lesions found, 14 false alarms

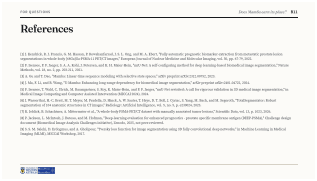
Case B was chosen by rule before I looked: the lowest F1 among scans where F1 is defined. It is not the worst scan, because nine scans had no lesion found and so have no F1. Both of its lesions were found. Its 14 false alarms were 8 in bone and 6 in soft tissue, and none sat in the kidneys or bladder, which the organ task targets.



`Backup B10 · 64 words`

The threshold and size filter were fixed before the test

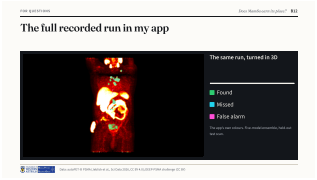
The operating point came from the five-fold validation sweep, before any test inference. With the 10-voxel filter, F1 moved by less than a point across thresholds, so I kept 0.5. The protocol fixed two rows: the filtered headline and an unfiltered row, because PSMASegmentator has no filter. On the test set the filter trades a little sensitivity for precision, and both rows are reported.



`Backup B11 · 27 words`

References

These are the works cited on the slides, in the order they appear, in the same IEEE style as the dissertation. The dissertation has the full list.



`Backup B12 · 62 words`

The full recorded run in my app

This is a real run of the five-model ensemble in the app I built, on a held-out test scan, sped up 6x. It needs a CUDA GPU, so I recorded it on my PC. The colours are the app's own: found, missed and false alarm. The code will be released so the UWA Medical Physics Research Group can retrain and score it.